

HERMANN PAUWELLS RIVERA

Software Engineer | C++ · Audio/DSP

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PROFILE

Software engineer focused on C++ and audio/DSP (JUCE), with full-stack experience in TypeScript, React/Next.js and Node.js, graduating soon from a Computer Science Engineering degree at Tecnológico de Monterrey. I built a VST3 synthesizer with automated tests and software for real businesses (e-commerce, real-time reservations). I also work with machine learning.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript/JavaScript, SQL, Swift, C#

Audio/Native: JUCE, VST3, DSP, CMake, pluginval

Web: React, Next.js, Node.js, Express, Supabase

Tools: Git, GitHub, GitHub Actions, CI, Unity

EXPERIENCE

Full-Stack Developer — Juguetería El Arbolito

2026–present

- Developing and implementing an e-commerce and admin platform for a single-branch retail business (catalog, cart, checkout, backend).
- Implementing Mercado Pago payments and inventory synchronization between online sales and the in-store point of sale.

SELECTED PROJECTS

folk-park

Personal project

2026

- Wavetable VST3/Standalone synthesizer in C++20 and JUCE; deterministic MIDI generation and real-time DSP architecture.
- Automated suites (16/16 suites, 19/19 UI contracts) and validation with pluginval strictness 5.

Lumina Reservations (WorkHub MTY)

Academic team project · Node/Express backend, real-time (SSE) and frontend

2026

- Implemented the reservation backend in Node/Express (services, repositories, controllers, migrations) and real-time occupancy updates over SSE.
- Built the floor map and the reservation flow in React/Vite (dashboard, filters, confirmation) with role controls.

HeatShield

iOS app

2025

Winner, Sustainability category, Swift Challenge Fest 2025 (Tecnológico de Monterrey)

- iOS app with real-time heat alerts (OpenWeatherMap), heat index and shelter maps.

Other projects: CrediFiel Challenge (Datathon 2025) · Multi-agent warehouse simulation (RL) · RainLab Northsky · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

EDUCATION

Computer Science Engineering (ITC) — Tecnológico de Monterrey, Monterrey Campus — 2021–2026 ·

Graduation expected: December 2026

ACTIVITIES

- SwiftTec Coding Community (2025) — hackathons and innovation challenges.
- Music production and sound design (FL Studio).

Spanish — native · English — C1